Canadian National Railway (TSX: CNR)

What is the franchise worth, and what is the share price assuming?

Matin Saiyed · Valuation memo · Price date 2026-11-10 · Companion workbook: CN_Valuation_Model.xlsx

How this was produced: the data pipeline, analysis, figures and text were produced with Claude, an AI model made by Anthropic, at the author's direction. The author set the brief, reviewed the output and is responsible for what is published. Code and data are public so every result can be re-run.

Summary. CN is one of the highest-quality freight franchises in North America, and its shares trade at a discount to other Class I railroads. A discounted cash flow model built from eight years of filed financials nevertheless values the equity at about **C$112 per share** in the base case, 38% below the C$179.80 market price, and at C$152 even in the upside case. Peer multiples point the other way, implying roughly C$210.

The two methods disagree because the whole sector is priced richly. To justify today's price the model needs either perpetual growth of 4.4% a year, a cost of capital of 6.0%, or five years of 13.1% annual revenue growth. My conclusion is that CN is **cheap relative to peers but expensive relative to its own cash flows**, and that the gap comes down to how much of its earnings must be reinvested in the network.

C$112	C$179.80	7.66%	15.4x vs 17.6x
DCF value per share, base	Share price	WACC	EV/EBITDA, CN vs peer median

1. The business

CN operates a rail network connecting Canada's Atlantic and Pacific coasts with the U.S. Gulf Coast. Freight railroads are natural duopolies or monopolies on most lanes: track cannot be replicated economically, and for bulk commodities over long distances there is no competitive substitute. That shows up as pricing power and high margins. It also shows up as heavy, non-discretionary capital spending, because the network has to be maintained whether volumes grow or not.

From FY2018 to FY2025 revenue grew from C$14,321m to C$17,304m, a compound rate of 2.7%. The operating ratio, operating expenses as a share of revenue and the industry's standard efficiency measure, was 61.9% in FY2025 (Figure 1). Diluted earnings per share rose from C$5.87 to C$7.57, faster than net income, because the diluted share count fell from 738m to 624m.

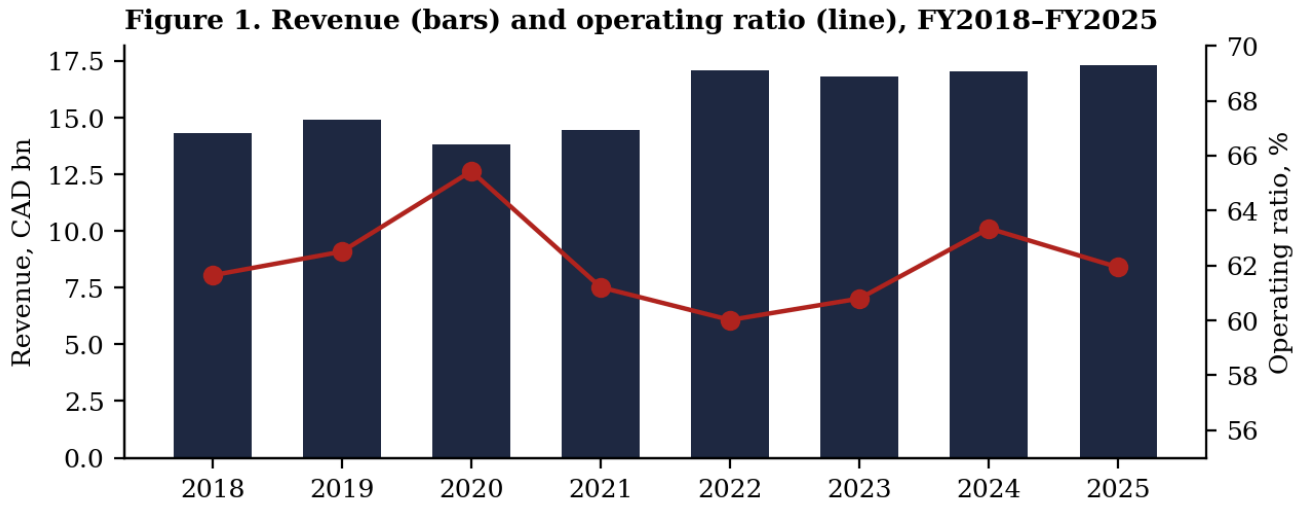
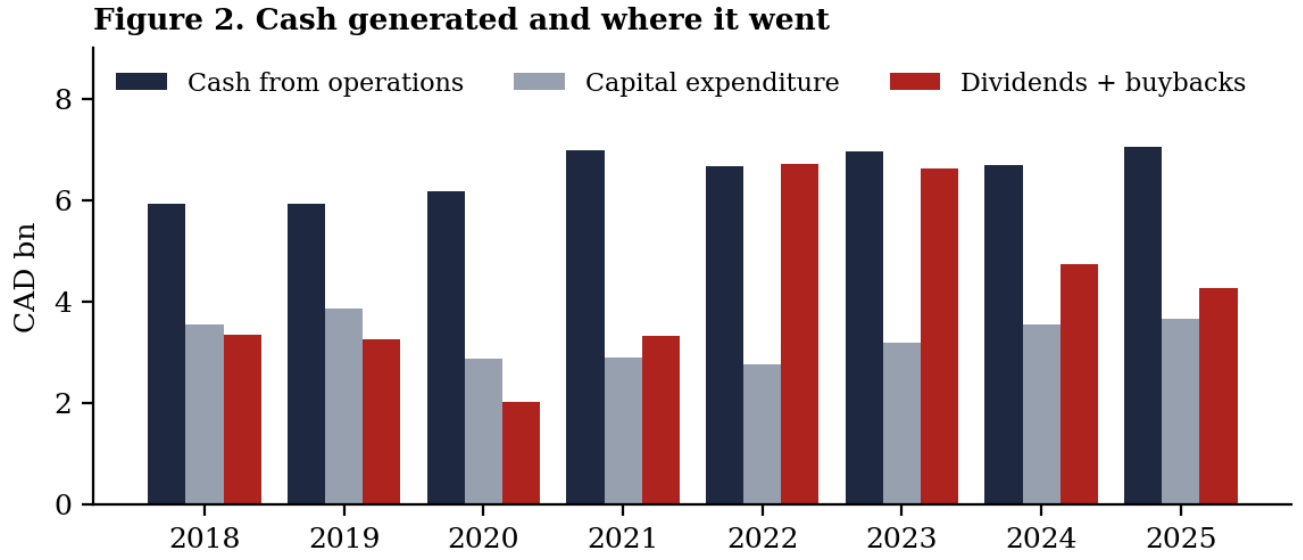


Figure 2 is the most important chart in this memo. Over eight years CN generated C$26.1bn of free cash flow after capital expenditure and returned C$34.3bn to shareholders through dividends and buybacks. The difference was funded with debt, which rose from C$12,569m to C$21,206m. Capital expenditure has averaged 19.4% of revenue over the last five years against depreciation of 11.2%. Reported earnings therefore overstate the cash an owner can take out.



2. Forecast

The forecast is driven by nine inputs, set out on the Inputs sheet of the workbook. In the base case revenue grows 4.0% a year, about two points of volume and two of price, and the operating ratio improves to 60.0% by 2030. Capital expenditure stays at its five-year average share of revenue, debt is held at 2.5 times EBITDA, 47% of net income is paid as dividends, and all remaining cash repurchases shares. The forecast balance sheet balances in every year by construction, which the workbook checks.

Table 1. Base-case forecast

Base case, CAD m	2026E	2027E	2028E	2029E	2030E
Revenue	17,996	18,716	19,465	20,243	21,053
Operating ratio	61.5%	61.2%	60.8%	60.4%	60.0%
Operating income (EBIT)	6,920	7,269	7,635	8,019	8,421
EBITDA	8,936	9,365	9,815	10,286	10,779
Net income	4,975	5,202	5,443	5,696	5,961
Earnings per share (CAD)	8.19	8.73	9.31	9.93	10.58
Capital expenditure	3,491	3,631	3,776	3,927	4,084
Unlevered free cash flow	3,735	3,939	4,154	4,379	4,615
Share repurchases	2,280	2,282	2,398	2,520	2,649

3. Cost of capital

The cost of equity is 8.50%, from a risk-free rate of 3.77% (Government of Canada 10-year benchmark), an equity risk premium of 5.0%, and a beta of 0.95. The beta is a 60-month regression against the S&P/TSX Composite (0.92), adjusted toward one. The pre-tax cost of debt is 4.34%, FY2025 interest expense over average debt. At market-value weights the weighted average cost of capital is **7.66%**.

4. Discounted cash flow

Free cash flows for FY2026–FY2030 are discounted with a mid-year convention, and a Gordon-growth terminal value assumes 2.5% perpetual growth. Enterprise value is C$83,058m. Subtracting C$21,206m of debt, adding C$350m of cash and C$3,706m for the after-tax pension surplus gives equity of C$65,908m, or C$107 per share at the balance sheet date and C$112 rolled forward to the price date. Terminal value is 79% of enterprise value, which is typical for a long-lived infrastructure asset and is the reason the sensitivity tables matter more than the point estimate.

Table 2. Scenarios

Driver	Downside	Base	Upside
Revenue growth per year	2.0%	4.0%	5.5%
Operating ratio by 2030	62.5%	60.0%	58.5%
Capex, % of revenue	20.4%	19.4%	18.4%
Terminal growth	2.0%	2.5%	3.0%
Enterprise value (CAD m)	61,860	83,058	105,597
Value per share (CAD)	75	112	152
Versus share price	−58%	−38%	−16%
Terminal value, % of EV	76%	79%	82%
Implied exit EV/EBITDA	6.9x	8.3x	9.8x

Figure 4. Forecast unlevered free cash flow by scenario

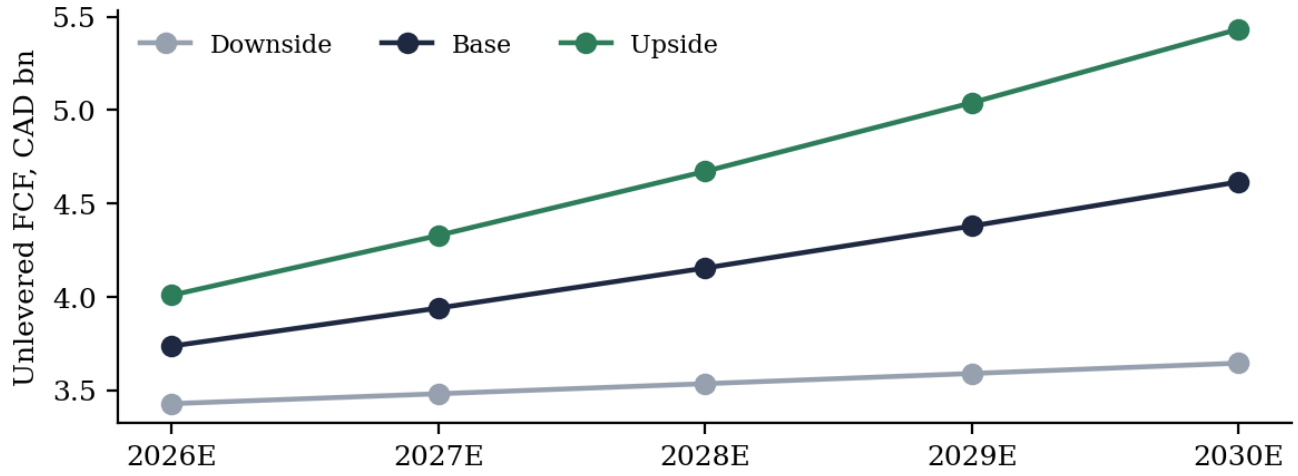


Table 3. Value per share (CAD): WACC and terminal growth

WACC \ terminal growth	1.5%	2.0%	2.5%	3.0%	3.5%
6.66%	117	130	147	168	196
7.16%	104	114	128	144	165
7.66%	93	102	**112**	125	141
8.16%	83	91	100	110	123
8.66%	75	82	89	98	108

Table 4. Value per share (CAD): operating ratio and revenue growth

2030 operating ratio \ revenue growth	2%	3%	4%	5%	6%
58%	109	115	121	128	135
59%	104	110	117	123	130
60%	100	106	**112**	119	125
61%	96	102	108	114	120
62%	92	97	103	109	115

No cell in either table reaches the current share price except the most favourable corner of Table 3. Table 4 shows that even a 58% operating ratio with 6% growth, well beyond anything in CN's eight-year record, yields C$135.

5. Comparable companies

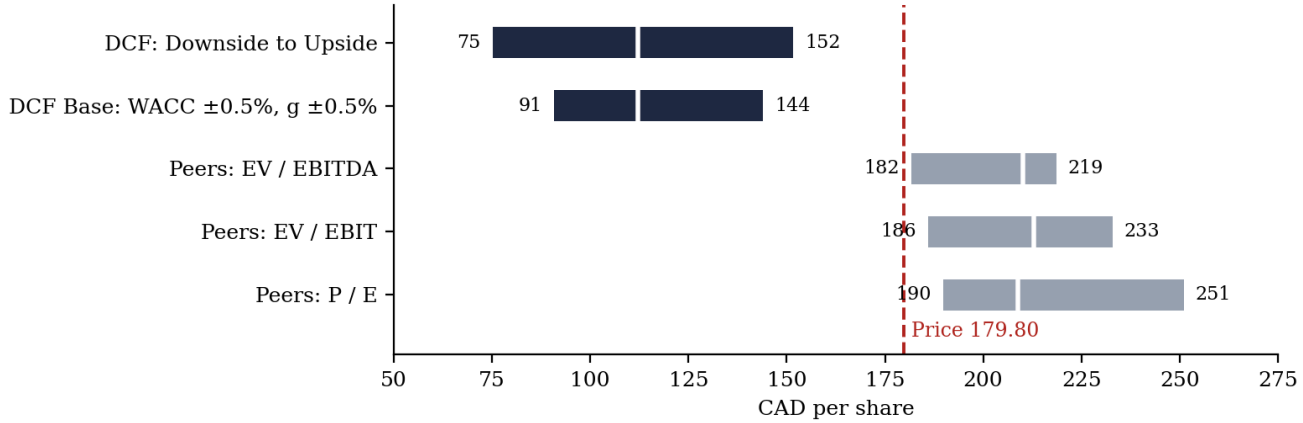
The four other publicly traded Class I railroads trade at a median of 17.6 times FY2025 EBITDA and 27.6 times earnings. CN trades at 15.4 and 23.8 times, a discount of 12% on EV/EBITDA despite an operating ratio better than three of the four. Applying peer medians to CN's FY2025 results implies C$210–213 per share.

Table 5. Class I railroads, FY2025

Company	Ccy	Market cap (m)	Ent. value (m)	EV/EBITDA	EV/EBIT	P/E	Op. ratio
Canadian Pacific Kansas City	CAD	115,633	138,637	18.2x	24.7x	29.5x	62.8%
Union Pacific	USD	178,644	209,192	17.0x	21.2x	25.1x	59.8%
CSX	USD	94,161	112,364	18.1x	24.9x	33.2x	67.9%
Norfolk Southern	USD	73,659	89,216	15.5x	20.5x	25.7x	64.2%
Peer median				**17.6x**	**23.0x**	**27.6x**	
Canadian National	**CAD**	**110,271**	**131,127**	**15.4x**	**19.9x**	**23.8x**	**61.9%**

Local currency. Norfolk Southern is subject to an agreed merger with Union Pacific, so its price reflects deal terms. Source: SEC filings; prices at 2026-11-10.

Figure 3. Value per share by method (white tick = midpoint)



6. Reconciling the two answers

A relative valuation says CN is worth more than its price, and an intrinsic valuation says it is worth less. Both are correct on their own terms. Multiples only say that CN is cheaper than its peers, not that any of them are cheap. The reverse DCF makes the market's assumptions explicit. Holding everything else at the base case, the share price is justified by any one of:

- terminal growth of 4.4% a year forever, above long-run nominal GDP growth;
- a cost of capital of 6.0%, about 1.7 points below my estimate;
- revenue growth of 13.1% a year for five years, against 2.7% achieved since 2018.

None of these is impossible, and a combination of smaller moves in each would do it. Investors may reasonably view a railroad as a long-duration, inflation-protected asset deserving a lower discount rate than CAPM produces. But the model shows what is being paid for, and I would want a specific reason to believe it before paying.

7. Risks and limitations

- **Capital intensity is the swing factor.** If maintenance capex is closer to depreciation than to recent spending, base value rises materially. I use reported spending because CN does not disclose a

maintenance/growth split.

- **Macro and trade.** Volumes depend on commodity cycles, cross-border trade policy and labour disruptions, none of which is modelled explicitly.
- **Simplifications.** Deferred taxes are treated as cash taxes, which understates cash flow slightly. Debt is taken at book value. Peer multiples are not adjusted for pensions, leases or one-off items. The comparison uses trailing FY2025 results, not forward estimates.
- **Timing.** The balance sheet is at 31 December 2025 and the price is 10 months later; value is rolled forward at the cost of equity less dividends.

Sources: Canadian National Railway annual financial statements (US GAAP) as filed with the U.S. Securities and Exchange Commission; peer filings on SEC EDGAR; Bank of Canada; Yahoo Finance. This memo is an educational valuation exercise based on public information. It is not investment advice or a recommendation to buy or sell any security, and the views are the author's own and not those of any employer.